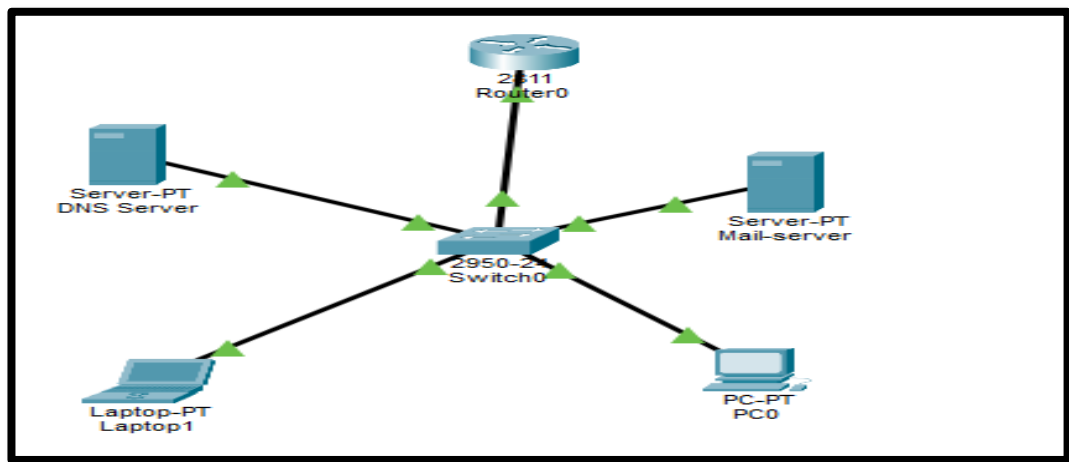


TP 3 : SMTP

Last Name : Belkacem

First Name : Nacim

GRP : 1 SGR : 1



Introduction :

This lab report presents the work completed during Lab 03, which focuses on the SMTP (Simple Mail Transfer Protocol). The objective of this lab is to study, configure, and evaluate a complete email system within a local network using Cisco Packet Tracer as a simulation platform. Email communication is a fundamental service in modern networks. Its functionality is based on the interaction of several important protocols:

- **SMTP (Simple Mail Transfer Protocol):** used for sending emails. It operates on port 25 for unencrypted communication and port 465 for secure transmission. It ensures the transfer of messages from the client to the mail server, as well as between mail servers.
- **POP3 (Post Office Protocol version 3):** used for receiving emails. It enables users to retrieve messages stored on the mail server.
- **DNS (Domain Name System):** a system used for resolving domain names into IP addresses. It plays a crucial role in email communication by allowing devices to locate servers.
- **DHCP (Dynamic Host Configuration Protocol):** automatically provides network configuration parameters such as IP address, subnet mask, gateway, and DNS server, avoiding manual setup.

During this lab, a complete network topology was implemented. The required services were configured, email communication was tested, the impact of disabling certain protocols was analyzed, and a DHCP service was configured on a Cisco 2811 router.

Devices and IP Addressing

The following table summarizes the devices used in the network along with their roles and IP addressing information:

Device	Role	IP Address	Subnet Mask
DNS Server (Server-PT)	Domain name resolution	192.168.1.11	255.255.255.0
Mail Server (Server-PT)	Email services (SMTP and POP3)	192.168.1.10	255.255.255.0
Laptop0 (user1)	Email client 1	192.168.1.2	255.255.255.0
PC0 (user2)	Email client 2	192.168.1.3	255.255.255.0
Switch0 (2950-24)	Layer 2 connectivity	N/A	N/A
Router0 (2811)	Default gateway and DHCP server	192.168.1.1	255.255.255.0

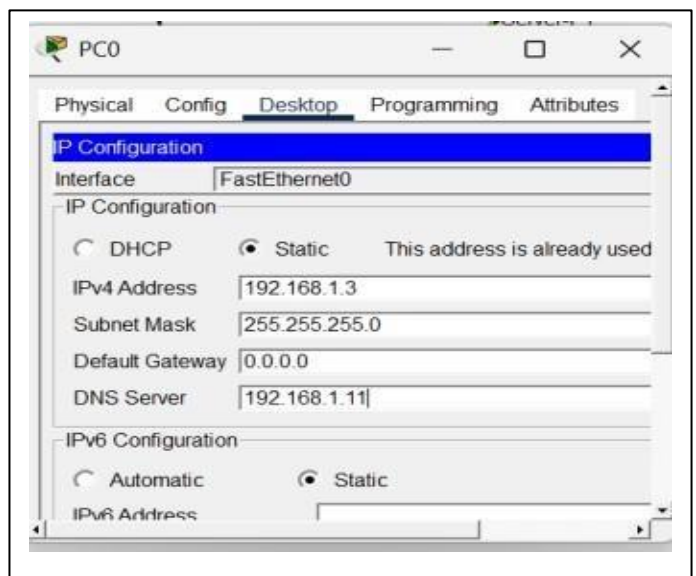
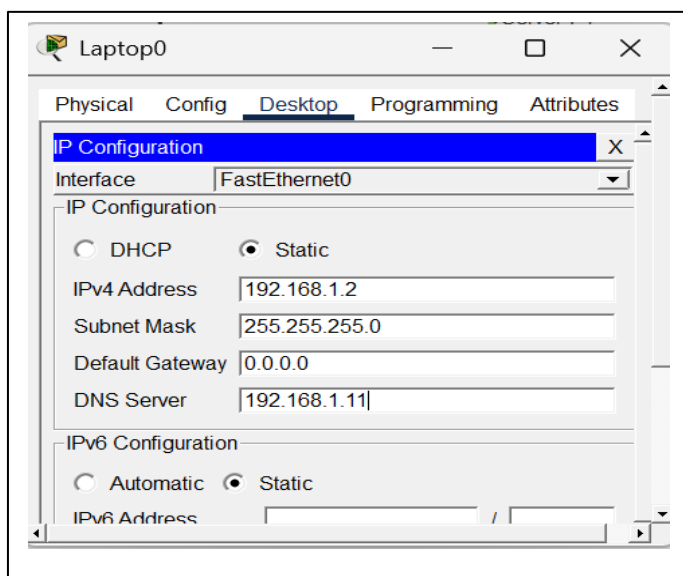
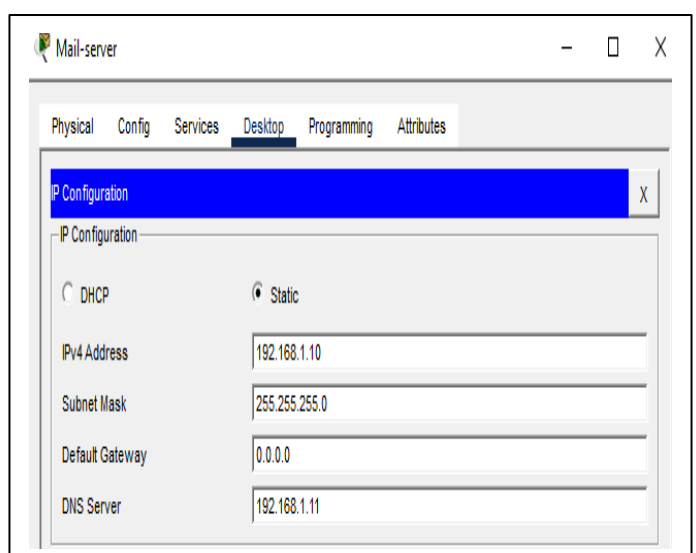
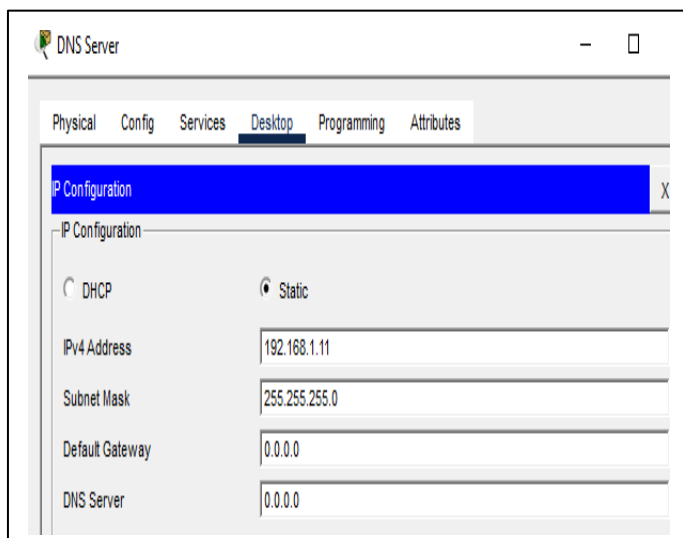
Building the Topology and Configuring Static IP Addresses :

Configuration Performed :

Each device was manually configured with a static IP address using the following path:

Desktop → IP Configuration → Static

The configuration details are shown below:



Configuring the Mail Server (SMTP + POP3) :

Explanation

The Mail Server plays a central role in the email system. It provides two main services:

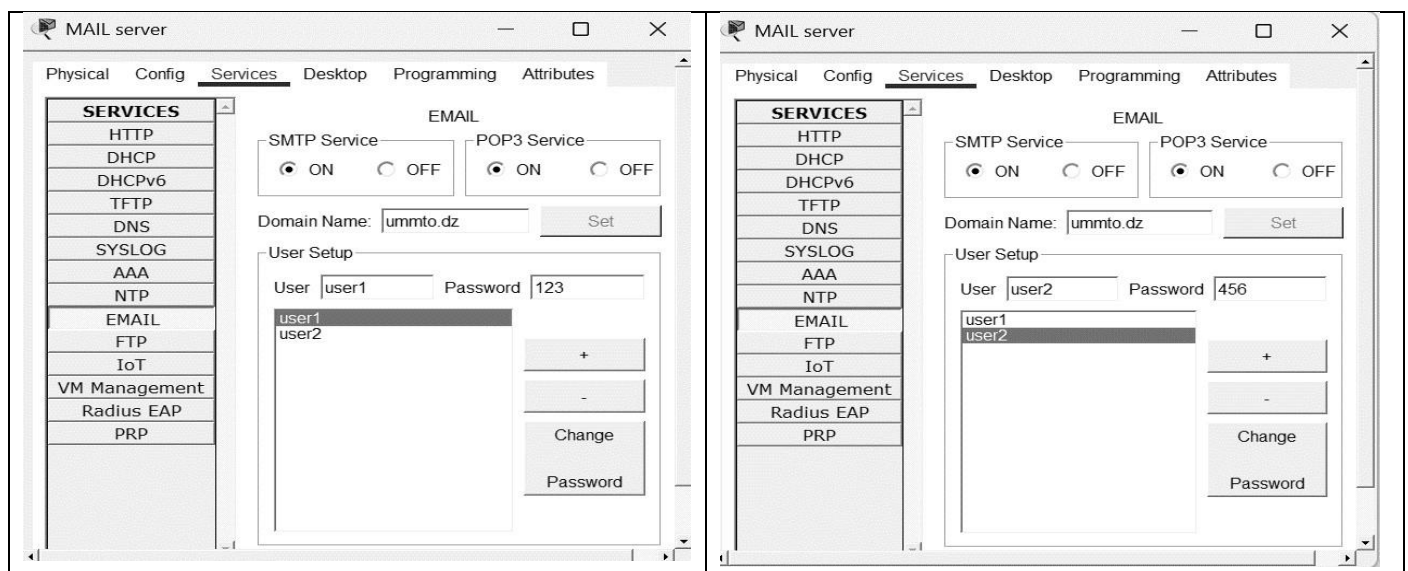
- **SMTP service:** responsible for receiving emails sent by clients and forwarding them to the appropriate destination. Without this service, sending emails is not possible.
- **POP3 service:** responsible for storing incoming emails and allowing clients to access and download them. Without this service, receiving emails is not possible.

The domain name configured on the server represents its identity. All user email addresses are based on this domain and follow the format: **username@domain_name**.

Configuration Procedure

1. Select the **Mail Server** in Cisco Packet Tracer.
2. Open the **Services** tab.
3. Choose **EMAIL** from the left panel.
4. Activate both services: set **SMTP = ON** and **POP3 = ON**.
5. Enter the domain name: **ummto.dz**, then click **Set**.
6. Create the following user accounts:

Username	Password	Full Email Address
user1	123	user1@ummto.dz
user2	456	user2@ummto.dz



Configuring the DNS Server :

Explanation

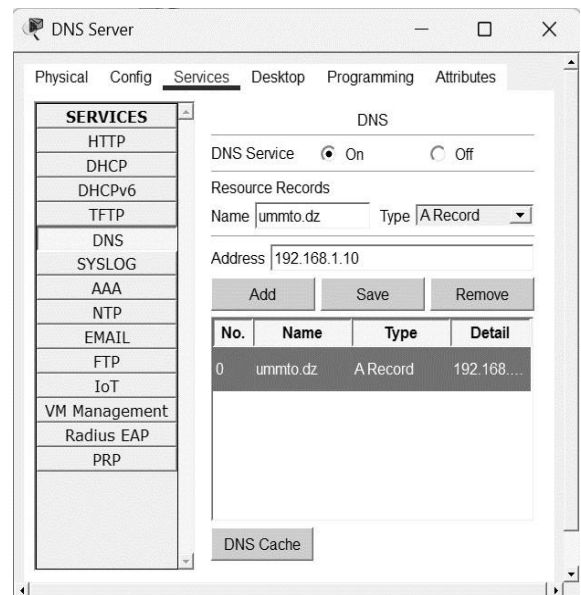
The DNS (Domain Name System) is an essential service in a computer network. It functions as a directory by converting human-readable domain names (such as *ummto.dz*) into IP addresses (such as *192.168.1.10*) that devices can understand.

Without DNS, network devices are unable to identify the IP address associated with the mail server *ummto.dz*, which would prevent the email system from operating correctly.

Configuration Procedure

7. Select the **DNS Server**.
8. Navigate to **Services → DNS**.
9. Enable the DNS service by setting it to **ON**.
10. Add the following entry:

Field	Value	Description
Name	ummto.dz	Domain name to be resolved
Type	A Record	Defines an IPv4 address mapping
Address	192.168.1.10	IP address of the Mail Server



DNS Resolution Test (ping ummto.dz) :

```
C:\ ping ummto.dz

Pinging 192.168.1.10 with 32 bytes of data:
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

Analysis

The results show that:

- The DNS server successfully resolves *ummto.dz* into the IP address *192.168.1.10*.
- The Mail Server is reachable from Laptop0, as indicated by the absence of packet loss.
- The network configuration of Laptop0 is correct, especially the DNS server address (*192.168.1.11*).

If the DNS address had not been configured on Laptop0, the command would have failed with the message:

"Ping request could not find host ummto.dz. Please check the name and try again."

Configuring Email Accounts on the Host Machines

Explanation

To send and receive emails, each user must configure their email client with the appropriate account information and server addresses.

- **Incoming Mail Server:** POP3 server used to retrieve emails.
- **Outgoing Mail Server:** SMTP server used to send emails.

In this configuration, both services are provided by the same server: *ummto.dz* (192.168.1.10).

Laptop0

Physical Config **Desktop** Programming Attributes

Configure Mail

User Information

Your Name: user1

Email Address: user1@ummto.dz

Server Information

Incoming Mail Server: ummto.dz

Outgoing Mail Server: ummto.dz

Logon Information

User Name: user1

Password: ●●●

Save Remove Clear Re

PC0

Configure Mail

User Information

Your Name: user2

Email Address: user2@ummto.dz

Server Information

Incoming Mail Server: ummto.dz

Outgoing Mail Server: ummto.dz

Logon Information

User Name: user2

Password: ●●●

Save Remove Clear Re

Top

Sending and Receiving an Email (Normal Test)

Explanation

This step validates the full operation of the email system from sender to receiver. The process follows these stages:

1. user1 composes and sends an email.
2. The SMTP client establishes a connection to the Mail Server (port 25).
3. The server uses DNS to resolve the recipient address.
4. The email is stored in user2's mailbox.
5. user2 retrieves the message using POP3.

PC0

MAIL BROWSER

Mails

Compose Reply Receive Delete onfig

	From	Subject	Re
1	user1@u...	test	Sun Apr 14:48:09

test

user1@ummto.dz

Sent : Sun Apr 5 2026 14:48:09

Hello user2

Top

Test with SMTP Service Disabled

Explanation

This test demonstrates the essential role of the SMTP protocol in sending emails. When the SMTP service on the Mail Server is disabled, clients are unable to establish a connection to send messages.

With SMTP turned off, the server immediately terminates any connection attempt on port 25 by sending a TCP **RST (Reset)** signal to the client.

Procedure

1. On the Mail Server: **Services → EMAIL → SMTP: OFF**
2. From **Laptop0**, attempt to send an email to *user2@ummtto.dz*
3. Observe the resulting error message

Result Obtained

(Connection attempt failed – error displayed)

Analysis

The error "*Connection received a reset from server*" indicates that the SMTP connection was rejected by the server. This confirms that:

- SMTP is indispensable for sending emails.
- Without an active SMTP service, the client cannot deposit messages on the server, even if POP3 is still functioning.
- SMTP is responsible for delivering emails from the sender to the recipient's mailbox on the server.

Test with POP3 Service Disabled

Explanation

This test illustrates the role of the POP3 protocol in receiving emails. By disabling POP3, clients cannot connect to the server to check their messages, even if the emails have already been successfully stored.

Procedure

1. On the Mail Server: **Services → EMAIL → SMTP: ON (re-enable), POP3: OFF**
2. From **Laptop0**, send a new email to *user2@ummtto.dz* (sending succeeds because SMTP is active)
3. From **PC0**, click **Receive** to attempt message retrieval
4. Observe the error message

Result Obtained

(Client cannot download messages – error displayed)

Analysis

This test highlights the distinction between SMTP and POP3:

- **SMTP active:** Emails can be sent and are stored on the server.
- **POP3 disabled:** Clients cannot retrieve stored messages.

The messages are physically present on the server but remain inaccessible until the POP3 service is re-enabled. Once POP3 is active again, the client can successfully download the stored messages.

Homework: Adding a Cisco 2811 Router as a DHCP Server

Configuring the DHCP Server via CLI :

```
Router> enable
Router# configure terminal
Router(config)#int fa0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no sh
Router(config-if)#exit

Router(config)# ip dhcp excluded-address 192.168.1.1 192.168.1.20

Router(config)# ip dhcp pool Mypool

Router(dhcp-config)# network 192.168.1.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.1.1

Router(dhcp-config)# dns-server 192.168.1.11

Router(dhcp-config)# exit
Router(config)# exit
```

Testing DHCP on Host Machines



Conclusion

This lab combined theory and practice to explore email protocols in a simulated network using Cisco Packet Tracer.

The study highlighted the **interdependence of key protocols**: DNS resolves domain names, SMTP handles sending messages, and POP3 allows clients to retrieve emails. Each is essential for proper email operation.

The exercise also emphasized the **importance of accurate configuration**. Mistakes like using the wrong domain name or omitting the DNS server address can completely disrupt communication.

Additionally, setting up **DHCP** on the Cisco 2811 router showed how automation simplifies network management by assigning unique IP addresses and reducing manual errors.

Overall, the lab provided a clear, practical understanding of email communication—from composing and sending messages to receiving them—while demonstrating the interaction between network protocols in a controlled environment.